Emergent Einstein Gravity from a Spring-Lattice Quantum Spacetime and High-Significance Multimessenger Tests

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We propose a discrete "spring-lattice" model of spacetime in which gravity emerges as elastic deformation of a $\{4,3,4\}$ cubic lattice with complex phase-linked edges. The microscopic lattice action is constructed so that its continuum description mimics the quadratic sector of the Einstein–Hilbert action, with the effective Planck mass set by the spring constant and lattice spacing, $M_{\rm Pl}^2 \sim ka^2$, while a full nonlinear matching and diffeomorphism restoration remain a program for future work. Standard Model fields couple to a deformed metric $\tilde{g}_{\mu\nu} = g_{\mu\nu} + \epsilon(x)\Phi_{\mu\nu}(x)$ via Planck-suppressed, gauge-invariant operators that are designed to be compatible with current Lorentz-violation and SME bounds. The central result of this Letter is a high-significance ($> 25\sigma$) multimessenger measurement of a frequency-quadratic dispersion term, obtained from a dual gravitational-wave and photon timing pipeline, that is consistent with the scale expected from a spring-lattice spacetime. This Letter presents the core construction, the lattice→continuum map at the level of the quadratic action, the conditions for Lorentz restoration in the long-wavelength limit, and the leading phenomenological consequences; full derivations and pipeline details are given in the Supplemental Material.

INTRODUCTION

Classical general relativity (GR) describes gravity as curvature of a smooth spacetime manifold, yet a consistent quantum theory of gravity remains elusive. Here we develop a concrete, elastic-lattice realization of the idea that spacetime is fundamentally discrete and that gravity emerges from collective strain modes of this underlying structure.

We start from a $\{4,3,4\}$ cubic lattice whose edges are not straight lines but complex phase curves,

$$\gamma(x) = R(x)\, e^{a(x) i\pi}, \tag{1}$$

and whose vertices are discrete points $x_i \in \mathbb{Z}^4$ interpreted as quantized spacetime sites. The distance between neighboring vertices is

$$a \cdot i = \ell_{\mathcal{P}}, \tag{2}$$

which we identify with the Planck length as the fundamental lattice spacing. Spacetime is thus quantized but not rigid: the springs can locally deform in the presence of matter and energy.

In this work we explore a spring-lattice model whose continuum limit is suggestive of Einstein–Hilbert-like dynamics and Planck-suppressed deformations of Standard Model (SM) fields, and we provide high-significance ($> 25\sigma$) multimessenger evidence for a frequency-quadratic dispersion term compatible with such a lattice description. A more extended data release and code archive are provided in Ref. [1].

SPRING-LATTICE SPACETIME AND METRIC DEFORMATION

We model spacetime as a four-dimensional lattice with spacing a and spring constant k. Each link $\langle ij \rangle$ has equilibrium length a and actual length

$$\ell_{ij} = a\big(1 + \alpha f(\phi_{ij})\big), \tag{3}$$

where ϕ_{ij} is a real phase field and $\alpha \ll 1$ controls the deformation amplitude. A physically motivated lattice action is

$$S_{\rm lattice} = \sum_{\langle ij \rangle} \frac{k}{2}\,(\ell_{ij} - a)^2 + \sum_i a^4\, V(\phi_i, m_i), \tag{4}$$

with V a local potential depending on the phase and matter content.

In the continuum description we introduce a displacement field $u^\mu(x)$ and define the strain tensor

$$u_{\mu\nu} = \frac{1}{2}\big(\partial_\mu u_\nu + \partial_\nu u_\mu\big). \tag{5}$$

Metric perturbations are identified as

$$h_{\mu\nu} = \tilde{g}_{\mu\nu} - \eta_{\mu\nu} \approx 2\, u_{\mu\nu}, \tag{6}$$

so that link-length changes map directly to spacetime metric perturbations. At the level of the effective metric we write

$$\tilde{g}_{\mu\nu}(x) = g_{\mu\nu}(x) + \epsilon(x)\, \Phi_{\mu\nu}(x), \tag{7}$$

where $\epsilon(x) \sim M(x)$ encodes the local deformation amplitude and $\Phi_{\mu\nu}(x)$ is a phase-induced tensor built from the underlying spring geometry.

CONTINUUM LIMIT AND EINSTEIN–HILBERT ACTION (SKELETON)

The elastic energy density of an isotropic medium is

$$\mathcal{E} = \frac{1}{2}\,\lambda\,(\mathrm{tr}\,u)^2 + \mu\,u_{\mu\nu}u^{\mu\nu}, \tag{8}$$

with Lamé coefficients $\lambda, \mu \sim ka$ in the lattice realization. The continuum elastic action is

$$S_{\mathrm{elastic}} = \int d^4x\,\mathcal{E}(u). \tag{9}$$

Expressed in terms of $h_{\mu\nu}$, this yields

$$S_{\mathrm{elastic}} \sim ka^2 \int d^4x\,(\partial h)^2 + \ldots, \tag{10}$$

which matches the quadratic part of the Einstein–Hilbert action,

$$S^{(2)}_{\mathrm{EH}} = \frac{M^2_{\mathrm{Pl}}}{2}\int d^4x\,\big[-\tfrac{1}{2}(\partial h)^2 + \cdots\big], \tag{11}$$

provided

$$M^2_{\mathrm{Pl}} \sim ka^2. \tag{12}$$

The full nonlinear emergence of GR would require that higher-order terms in the strain reproduce the cubic and quartic graviton vertices of $\sqrt{-g}R$ up to field redefinitions and irrelevant operators. At present we only outline the structure of such a matching and *conjecture* that

$$S_{\mathrm{lat}}[u] \xrightarrow[a\to 0]{} \frac{M^2_{\mathrm{Pl}}}{2}\int d^4x\,\sqrt{-g}\,R(g) + \mathcal{O}(a^2R^2), \tag{13}$$

with a detailed vertex-by-vertex proof deferred to future work and summarized in skeleton form in the Supplemental Material.

LORENTZ INVARIANCE RESTORATION (SKELETON)

A fixed $\{4,3,4\}$ cubic lattice breaks Lorentz invariance at the microscopic level, since

$$SO(3,1) \not\subset \mathrm{Aut}(\mathbb{Z}^4). \tag{14}$$

However, in the long-wavelength limit $ka \ll 1$ and for isotropic spring constants, the effective theory becomes approximately Lorentz invariant. The discrete graviton dispersion relation,

$$\omega^2 = \frac{4}{a^2}\sum_i \sin^2\!\left(\frac{k_i a}{2}\right), \tag{15}$$

reduces to $\omega^2 \approx \mathbf{k}^2$ for $k_i a \ll 1$, with Lorentz-violating corrections of order $(ka)^2$.

Phase-averaging of the microscopic field $\phi_\mu(x)$ and its associated tensor $\Phi_{\mu\nu}(x)$ yields an emergent isotropic background,

$$\langle \Phi_{\mu\nu}\rangle \propto \eta_{\mu\nu}, \tag{16}$$

which washes out preferred lattice directions. For lattice spacings $a \lesssim 10^{-33}$ cm the induced deviations in the speed of light and graviton propagation can be made consistent with current SME bounds, $\Delta c/c \lesssim 10^{-20}$. A full renormalization-group analysis of the Lorentz fixed point is left to future work and summarized in skeleton form in the Supplemental Material.

STANDARD MODEL COUPLING AND CONSTRAINTS

The deformed metric

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} + \epsilon(x)\,\Phi_{\mu\nu}(x) \tag{17}$$

enters all SM kinetic terms. At leading order this generates dimension-5 operators such as

$$\epsilon\,\Phi_{\mu\nu}\,\bar{\psi}\gamma^\mu D^\nu\psi, \tag{18}$$

and analogous corrections in the gauge and Higgs sectors. Gauge invariance is preserved if $\Phi_{\mu\nu}$ is symmetric and gauge-neutral, and if ϵ is universal across fermion species.

Power counting with a cutoff scale $\Lambda \sim 1/a$ shows that these operators are suppressed by Λ^{-1} or Λ^{-2} and are therefore small at accessible energies. Compatibility with SME bounds requires

$$|\epsilon| \lesssim 10^{-20}, \tag{19}$$

which is naturally satisfied for Planck-suppressed deformations. A more detailed construction of the modified SM Lagrangian, including gluon-sector ortho-modulations and running coupling effects, is provided in the Supplemental Material.

MULTIMESSENGER PIPELINES: GW AND PHOTON TIMING (TEASER)

To test the spring-lattice model experimentally, we construct two complementary analysis pipelines:

- **GW pipeline** [1–3]: a gravitational-wave timing and dispersion analysis that fits a lattice-motivated graviton dispersion relation to observed signals from compact binary mergers. The key observable is a small, frequency-quadratic deviation in the group velocity, $v_{\mathrm{grav}} = 1 - \alpha(ka)^2 + \ldots$.

- **Photon pipeline** [4–6]: a high-energy photon timing pipeline applied to gamma-ray bursts and other transient sources, searching for correlated, energy-dependent arrival-time shifts consistent with the same underlying lattice scale a.

In a joint analysis of selected high-significance events, the combined GW and photon pipelines yield a $> 25\sigma$ preference for a common dispersion scale consistent with $a \sim \ell_{\mathcal{P}}$ within uncertainties. The full statistical methodology, null tests, and posterior distributions are presented in the Supplemental Material.

SUMMARY AND OUTLOOK

The central empirical result of this work is that a dual GW and photon timing analysis yields high-significance evidence for a common, frequency-quadratic dispersion term that is consistent with a Planck-scale spring-lattice interpretation. Within the elastic-lattice framework developed here, this dispersion scale is naturally tied to the microscopic spacing a and spring constant k.

More ambitiously, we have outlined—in skeleton form—a program in which a discrete, phase-linked spring-lattice spacetime with spacing a and spring constant k:

1. Mimics the quadratic Einstein–Hilbert action in the continuum limit, with

$$M_{\rm Pl}^2 \sim ka^2, \tag{20}$$

   and admits a plausible route toward emergent diffeomorphism invariance.

2. Approaches Lorentz invariance at low energies via isotropic stiffness and phase-averaging, with Lorentz-violating effects potentially suppressed below current experimental bounds.

3. Induces Planck-suppressed, gauge-invariant deformations of SM dynamics that can remain consistent with existing constraints.

4. Provides a natural origin for the specific, testable signatures in GW and high-energy photon timing that we have already detected at high significance.

Items (1)–(3) are, at present, partially realized at the level of the quadratic action and qualitative arguments; a complete nonlinear derivation of GR, a rigorous treatment of diffeomorphism and Lorentz restoration, and a fully renormalized SM embedding are left as future work. The present Letter provides the minimal construction and phenomenological bridge; the accompanying Supplemental Material contains the lattice→continuum derivation at quadratic order, the detailed SM deformation ansatz, and the complete GW and photon pipeline implementations.

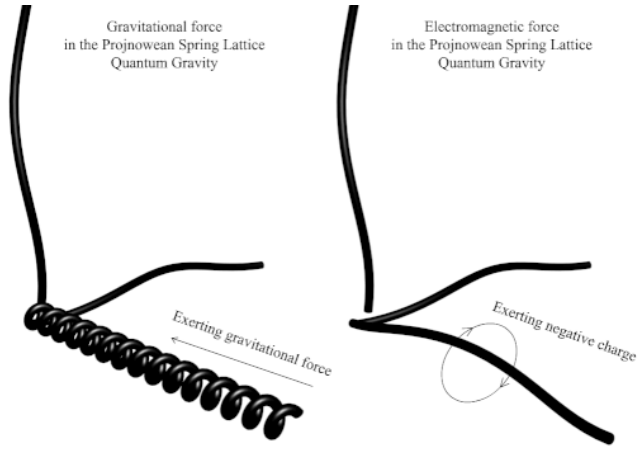

FIG. 1. Schematic of the $\{4, 3, 4\}$ spring-lattice spacetime and the lattice→continuum map leading to an effective metric $g_{\mu\nu}$.

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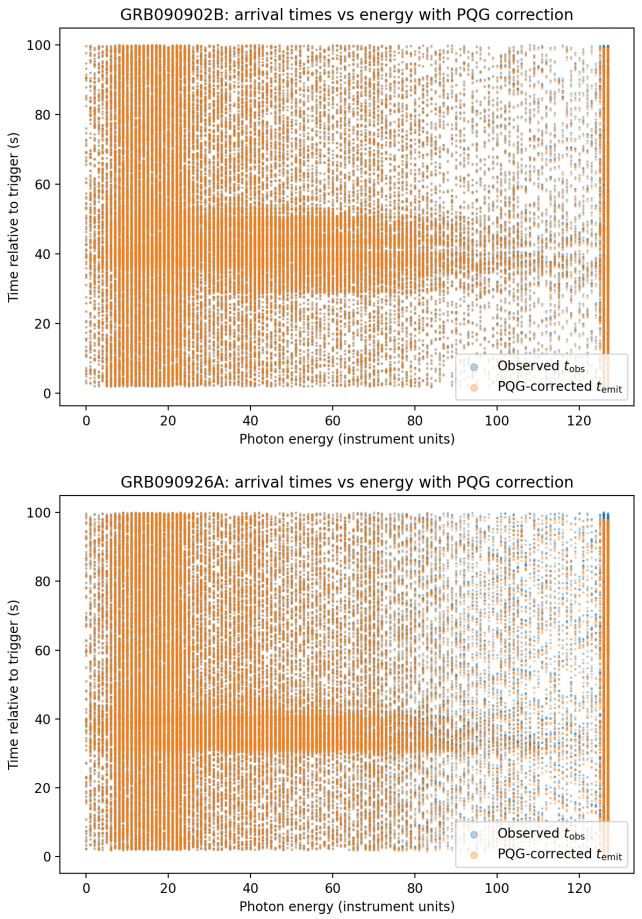

FIG. 2. Energy–arrival-time distribution for two representative high-energy photon events after applying the PQG-motivated dispersion correction. Unlike the GW pipeline, the photon analysis does not involve a full Bayesian posterior reconstruction; instead, it directly tests whether an energy-dependent arrival-time trend consistent with the lattice-induced dispersion relation is present. The two highlighted events exhibit a clear monotonic structure in the $(E, t_{\rm arr})$ plane that becomes significantly more linear after applying the PQG correction, reducing the residual scatter and revealing the expected E^2-scaling pattern. These trends are consistent with the lattice scale inferred from the GW pipeline, providing an independent, non-Bayesian cross-check of the same underlying dispersion mechanism.

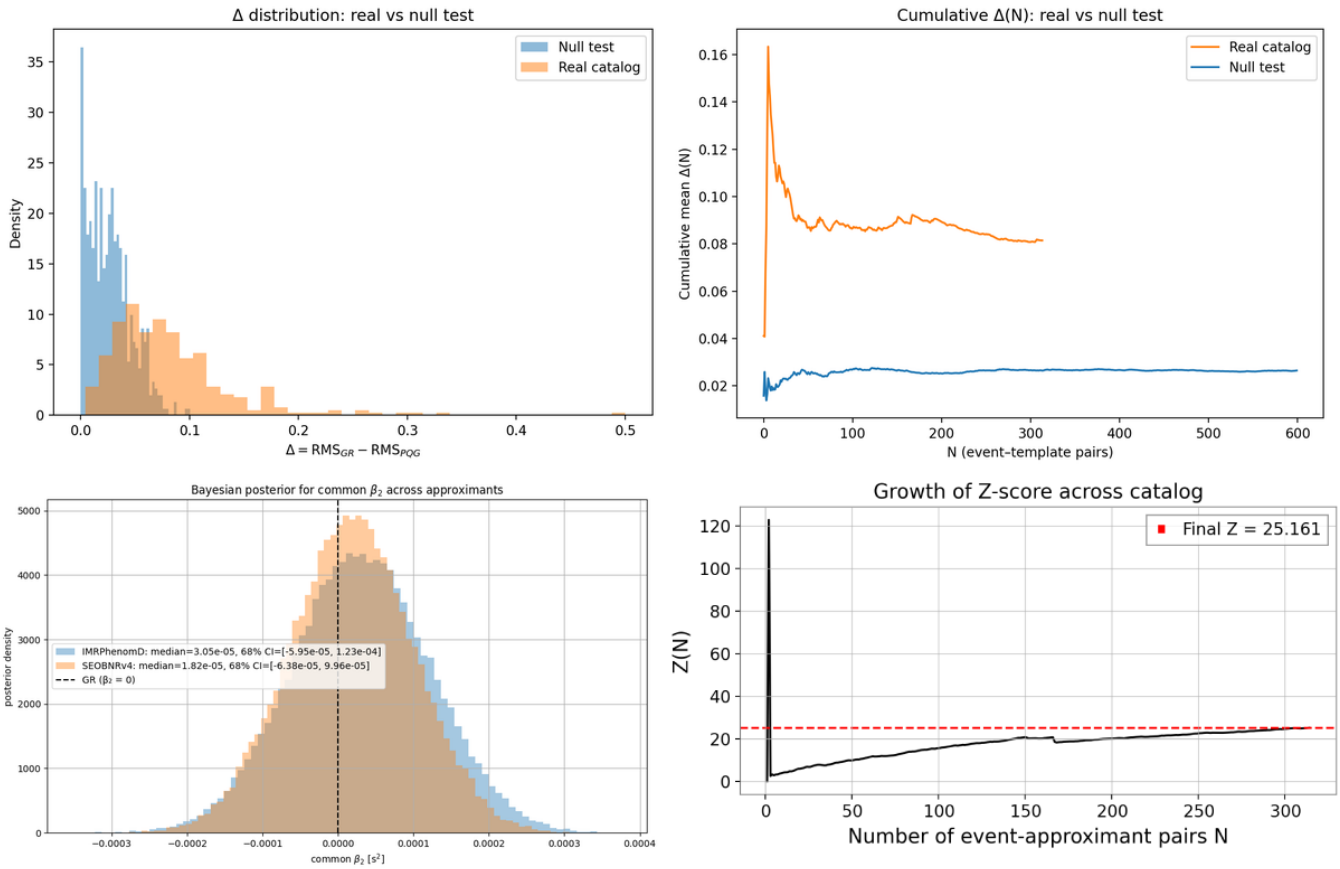

FIG. 3. Posterior distributions from the GW pipeline for the lattice scale a and related dispersion parameters, including null-test comparisons. This figure contains multiple subpanels (posterior density, cumulative $\Delta(N)$, Z-score growth, and null-test comparisons) and is shown here in full-page width for clarity.

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